Pygatha: MaxEnt constrained inference

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1 Outline

Suppose K players, with N cards. Each card must belong to one player or else be in the mystery pile,

$$\sum_{k}^{K+1} p_{k,n} = 1, \tag{1}$$

and each player must have M_k cards,

$$M_k = \lfloor \frac{N-3}{K} \rfloor + r_k. \tag{2}$$

$$\sum_{n=1}^{N} p_{k,n} = M_k. \tag{3}$$

Whilst playing the game we accumulate, C_k constraints for the k-th player,

$$\sum_{n=1}^{N} w_n^c p_{n,k} = 1 \tag{4}$$

Note that constraints 3 and 4 are both player constraints. This gives a formula for the entropy subject to Lagrangian constraints,

$$\mathcal{L} = \sum_{n,k} p_{n,k} \log p_{n,k} + \sum_{n} \lambda_n (\sum_{k} p_{n,k} - 1) + \sum_{k} \sum_{c=1}^{C_k} \mu_{k,c} \left(\sum_{n} w_{n,k}^c p_{n,k} - F_c \right). \tag{5}$$

We want to minimize this expression, Clearly,

$$\frac{\partial \mathcal{L}}{\partial p_{n,k}} = 1 + \log p_{n,k} + \lambda_n + \sum_c \mu_{k,c} w_{n,k}^c \tag{6}$$

When $\frac{\partial \mathcal{L}}{\partial p_{n,k}} = 0$,

$$p_{n,k} = \exp\left(-\lambda_n - \sum_c \mu_{k,c} w_{n,k}^c - 1\right), \tag{7}$$

Enforcing constraint 1, gives

$$\exp(-\lambda_n - 1) = \frac{1}{Z(\lambda, n)}, \tag{8}$$

where

$$Z(\lambda, n) = \sum_k^{K+1} \exp\left(-\sum_{c=1}^{C_k} \mu_{k,c} w_{n,k}^c\right). \tag{9}$$

So,

$$p_{n,k} = \frac{\exp\left(-\sum_c \mu_{k,c} w_{n,k}^c\right)}{Z(\lambda, n)}. \tag{10}$$

Now we formulate the dual problem for this $p_{n,k}$, the Lagrangian can be written,

$$\mathcal{D} = \sum_{n,k} p_{n,k} \left(\log p_{n,k} + \lambda_n + \sum_c \mu_{k,c} w_{n,k}^c\right) - \sum_n \lambda_n - \sum_k \sum_{c=1}^{C_k} \mu_{k,c} F_c. \tag{11}$$

and from, equation 6 at the minimum,

$$\mathcal{D} = -\sum_n (1 + \lambda_n) - \sum_k \sum_{c=1}^{C_k} \mu_{k,c} F_c. \tag{12}$$

$$\mathcal{D} = -\sum_n \log\left(Z(\lambda, n)\right) - \sum_k \sum_{c=1}^{C_k} \mu_{k,c} F_c. \tag{13}$$

We want to maximize this expression.

2 Computing the gradient

To plug this into ones favourite optimization algorithm, eg L-BFGS, we need a formula for the gradient of $\mathcal{D}$ with respect to $\mu_{k,c}$.

$$\nabla \mathcal{D}_{k,c} = \frac{\partial \mathcal{D}}{\partial \mu_{k,c}}, \tag{14}$$

$$\nabla \mathcal{D}_{k,c} = -\sum_n \frac{1}{Z(\lambda, n)} \frac{\partial Z}{\partial \mu_{k,c}} - F_c, \tag{15}$$

and

$$\frac{\partial Z}{\partial \mu_{k,c}} = \sum_{k'}^{K+1} \left(-\sum_{c'=1}^{C_k} \delta_{k,k'} \delta_{c,c'} w_{n,k'}^{c'} \right) \exp\left(-\sum_{c'=1}^{C_k} \mu_{k',c'} w_{n,k'}^{c'} \right), \tag{16}$$

$$= -w_{n,k}^{c} \exp\left(-\sum_{c'=1}^{C_k} \mu_{k,c'} w_{n,k}^{c'} \right) \tag{17}$$

So,

$$\nabla \mathcal{D}_{k,c} = \sum_n \frac{w_{n,k}^{c}}{Z(\lambda, n)} \exp\left(-\sum_{c'=1}^{C_k} \mu_{k,c'} w_{n,k}^{c'} \right) - F_c. \tag{18}$$

from equation 3 we see that,

$$\nabla \mathcal{D}_{k,c} = \sum_n w_{n,k}^{c} p_{n,k} - F_c. \tag{19}$$

Therefore, at the maximum,

$$\sum_n w_{n,k}^{c} p_{n,k} - F_c = 0, \tag{20}$$

and all the constraints are satisfied. We now have an update scheme for fitting $\mu_{k,c}$ to satisfy each players constraints.

3 Computing the Hessian

Clearly,

$$\nabla^2 \mathcal{D}_{(k,c),(k',c')} = -\sum_n \frac{w_{n,k}^c}{Z(\lambda,n)^2} \exp\left(-\sum_{c'=1}^{C_k} \mu_{k,c'} w_{n,k}^{c'}\right) \frac{\partial Z}{\partial \mu_{k',c'}} \tag{21}$$

$$+ \frac{w_{n,k}^c w_{n,k'}^{c'}}{Z(\lambda,n)} \delta_{k,k'} \exp\left(-\sum_{c'=1}^{C_k} \mu_{k,c'} w_{n,k}^{c'}\right) \tag{22}$$

Substituting,

$$E_{k,n} = \exp\left(-\sum_{c'=1}^{C_k} \mu_{k,c'} w_{n,k}^{c'}\right), \tag{23}$$

into this equation gives,

$$\nabla^2 \mathcal{D}_{(k,c),(k',c')} = -\sum_n \frac{w_{n,k}^c}{Z(\lambda,n)^2} E_{k,n} \frac{\partial Z}{\partial \mu_{k',c'}} + \frac{w_{n,k}^c w_{n,k'}^{c'}}{Z(\lambda,n)} \delta_{k,k'} E_{k,n} \tag{24}$$

$$\nabla^2 \mathcal{D}_{(k,c),(k',c')} = \sum_n \frac{w_{n,k}^c w_{n,k'}^{c'}}{Z(\lambda,n)^2} E_{k,n} E_{k',n} - \frac{w_{n,k}^c w_{n,k'}^{c'}}{Z(\lambda,n)} \delta_{k,k'} E_{k,n}. \tag{25}$$

$$\nabla^2 \mathcal{D}_{(k,c),(k',c')} = \sum_n w_{n,k}^c w_{n,k'}^{c'} p_{n,k} p_{n,k'} - w_{n,k}^c w_{n,k'}^{c'} \delta_{k,k'} p_{n,k}. \tag{26}$$

So,

$$x^T \left[\nabla^2 \mathcal{D}\right] x = \sum_n \left[\sum_{k,c} w_{n,k}^c p_{n,k} x_{k,c}\right]^2 - \sum_{k,c} w_{n,k}^c p_{n,k} x_{k,c} \sum_{c'} w_{n,k}^{c'} x_{k,c'}. \tag{27}$$

Let,

$$u_{n,k} = \sum_c w_{n,k}^c x_{k,c}, \tag{28}$$

then,

$$x^T \left[\nabla^2 \mathcal{D}\right] x = \sum_n \left[\sum_k p_{n,k} u_{n,k}\right]^2 - \sum_k p_{n,k} u_{n,k}^2. \tag{29}$$

Since, $\sum_k p_{n,k} = 1$, by the Cauchy Schwarz inequality

$$\left(\sum_k p_{n,k} u_k \right)^2 \leq \sum_k p_{n,k} u_k^2. \tag{30}$$

It follows that,

$$x^T \left[\nabla^2 \mathcal{D} \right] x \leq 0. \tag{31}$$

Therefore $\nabla^2 \mathcal{D}$ is positive-semi definite, and so the problem is convex.